THE CHEMICAL FACTORY

Built to Dominate

Chem Swell



Hydrophilic polymer watrstop for sealing construction joitns, concrete structure and critical waterproofing application.

FAST SWELLING

EXCELENT WATER SEALING

CHEMICAL RESISTANCE

EASY APPLICATION

DESCRIPTION

CHEM SWELL is an ultra-thin, profiled hydrophilic polymer waterstop designed for sealing construction and structural joints against water ingress. Its three-dimensional polymer structure is based on chemically resistant polyurethane polymer chains with hydrophilic functionality. On contact with water, the profile expands substantially and forms a swelling seal within the joint. The material is designed to accommodate repeated wet/dry cycles through controlled swelling and shrinking.

- Hydrophilic, re-swellable polymer waterstop for construction joints.
- Expands approximately 750–800% by volume on contact with water.
- Polymer structure designed to resist micro-bacterial attack.
- Suitable for water-retaining structures and underground construction applications.
- Capable of repeated swelling and shrinking during wet/dry cycles.

RECOMMENDE USES

- Sewerage treatment.
- Water tanks and reservoirs.
- Waste water treatment structure.
- Structure / Concrete pipes
- Swimming pools.



SERVICES

- Admixtures
- Surface Treatment
- Concrete Repair
- Water Proofings
- Industrial Flooring System
- Grouts & Anchors
- Adhesives
- Joint Sealants
- Bitumin Products
- Miscellaneous

STRONG WATERPROOFING

FAST & EFFECTIVE INJECTION

LONG LASTING PROTECTION

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TECHNICAL DATA

Property	Value
Product Type	Hydrophilic polymer waterstop / re-swellable joint sealing profile
Density	Approx. 1.0 g/cm³
Colour	Blue; other colours may be available on request
Swelling Retardation	Yes
Hardness	Approx. 3.8 Shore A
Tensile Strength	Approx. 1.5–2.1 MPa
Elongation	Approx. 490–770%
Hydrostatic Pressure	Approx. >50 m (5 bar)
Tear Strength	Swelling Capacity
Packaging	10 m roll / carton; other packing may be available
Profile Types	Rectangle and box profile
Available Sizes	5 × 20 mm; 10 × 20 mm

* The hardness value is reproduced from the supplied reference as stated. The reference identifies the value as approximately 3.8 (Shore A type); confirm the approved QC specification before issuing this figure as a certified laboratory value.

CONSUMPTION

Coverage: A swell bar, also known as a hydrophilic waterstop, usually goes along the whole length of a concrete joint and comes in rolls that are 5 to 40 meters long.

Guidence: A hydrophilic swell bar is a flexible waterstop strip used in construction joints to block water leaks by expanding when wet.

STANDARDS

- Solid Content: ~100%
- Hardness: Approx. 3.8 Shore A
- Density: Approx. 1.0 g/cm³

IMPORTANT INFORMATION

Supplied in: 10 meter roll / carton

Storage: Storage Condition Cool, covered and dry place; protect from direct sunlight and heat

Self Life: Shelf Life: 24 months in unopened original packaging, stored as suggested.

Hazard Class: Non-hazardous goods.

APPLICATION GUIDELINES

DISPENSING

Chem Swell (also called a hydrophilic waterstop or swellow bar) is a pre-formed rubber or polymer strip used in construction to seal concrete joints and stop water leaks.

MIXING

To use waterproof cement or mortar correctly, you must clean and dampen the surface, mix the product with the precise amount of water or additive, and apply at least two thin, alternating coats.

USES

Waterproofing Membrane on exposed flat and pitched roofs both for new construction and refurbish-ment of old roofs.

APPLICATION

To seal precast concrete elements, the given reference suggests creating a notch or recess where necessary. Choose a profile size so that the assembled elements create adequate compression /pre-stress at the joint. To fix the junction, use an appropriate mounting cement or permanently elastic sealant.

STORAGE & HANDLING

Chem Swell in its original box somewhere cool, protected, and dry. Keep the product away from direct sunlight, extreme temperatures, heat sources, and excess humidity. In tropical climates, air-conditioning is recommended for storage. High temperatures and humidity can cause degradation and shorten shelf life.

SAFETY

To ensure safety during installation, wear protective clothing, gloves, and goggles, avoid skin and eye contact, and seek medical advice if necessary. If accidentally swallowed, do not induce vomiting and seek medical attention immediately. Refer to the current Safety Data Sheet (SDS) for complete safety information.



ENHANCED DURABILITY

FASTER PLACEMENT

BETTER FINISH & PERFORMANCE

The information above reflects The Chemical Factory's current knowledge and is provided to assist specifiers and contractors. It does not replace project-specific trials to confirm suitability. Data are typical values under standard conditions; on-site results may vary. Recommendations are offered in good faith without warranty, as factors beyond our control can affect performance. The Chemical Factory may revise specifications without prior notice. Users are responsible for compliance with applicable standards and regulations. Technical assistance is available on request.



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